

AMCIS 2026 · RENO, NEVADA

Structural Biases in LLM-as-a-Judge Systems: Implications for reliable IS Adoption

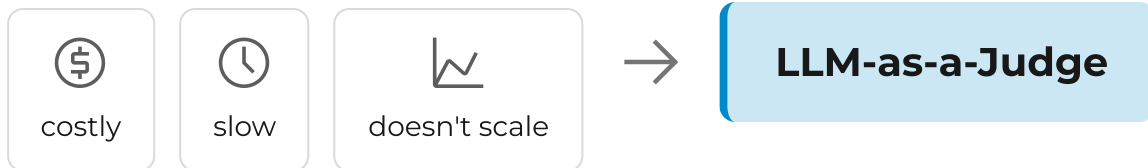
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BACKGROUND

What is LLM-as-a-Judge

human evaluation



already deployed today



Pre-screening applications



Assisting juries



Code review

Issues

- sensitivity to prompts and language
- variability depending on the task
- the existence of systematic biases, such as:

01

Self-preference bias

the judge favors the content it produced itself, or that of closely related models

02

AI-AI bias

the judge prefers machine-written content over human-written content

03

Position bias

the order in which the two candidates are shown changes the verdict

... and others

WHY IT MATTERS

One number to keep in mind:

23–60%

more likely to be shortlisted when the candidate used
the **same LLM as the recruiter**

Simulated hiring pipeline - Xu et al., 2025

Dictionaries as a Human Gold Standard

On subjective benchmarks, human answers vary too much to separate **judge bias** from **content quality**.

Professional

curated reference sources with
verified non-LLM provenance

Comparable

same task, same target format, so
sources sit side by side

Accessible

freely available entries

If strong biases appear even in this **narrow, controlled setting**, there is no reason to expect them to disappear in fuzzy organizational tasks.

RESEARCH QUESTION

To what extent do intrinsic biases - such as **AI-AI bias**, **self-preference** and **position** - pose risks to reliable acceptance of the LLM-as-a-Judge paradigm in organizational context?

METHOD

Corpus

100 English words

COCA frequency list

5 dictionaries → 500 human definitions

Cambridge, Collins, Merriam-Webster, Oxford ALD, Wiktionary

4 LLMs → 400 synthetic definitions

DeepSeek R1, GPT-5, Llama 3 8B (local), Mistral Large

Generation prompt - deliberately minimal:

"You are a dictionary editor. For each word received, produce a dictionary definition entry. Include: part(s) of speech, word senses with definitions, and usage examples where helpful."

METHOD

Evaluation Protocol

- **Blind pairwise comparison:** definition A vs B
- **Anonymized:** sources hidden from the judge
- **Randomized order:** A first or B first, 50/50

4 LLMs × 5 dicts → 20 pairs/word

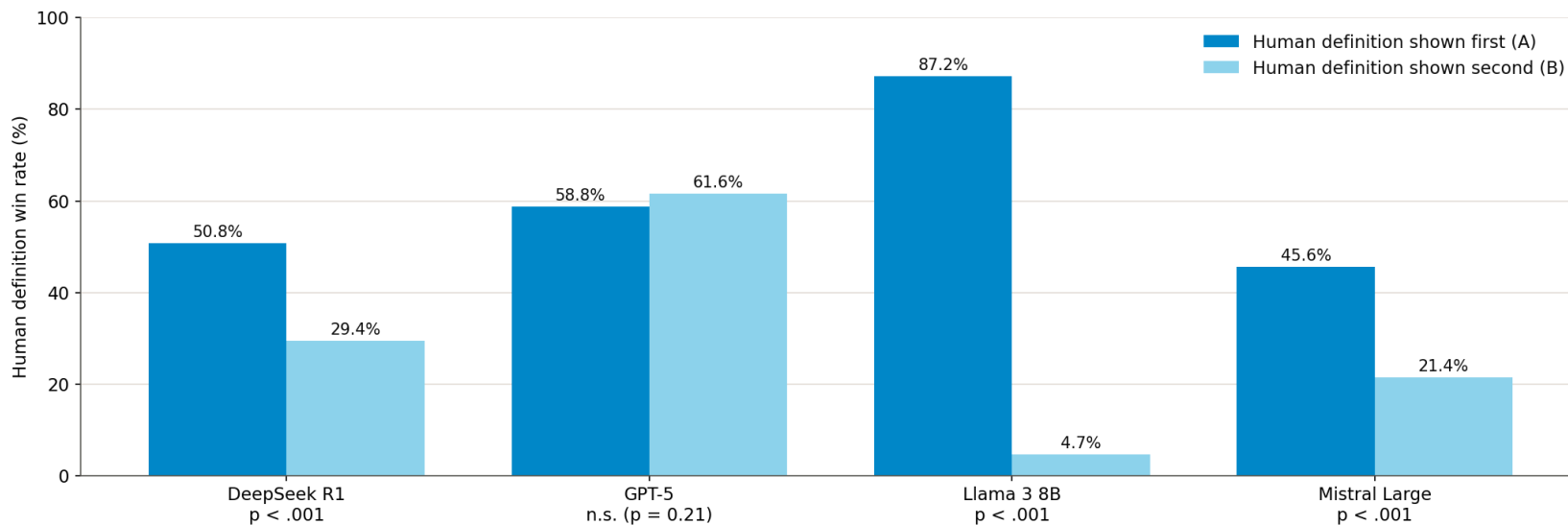
20 × 100 words → 2,000 pairs

2,000 × 4 judges → **8,000 judgments**

Judge prompt - role, no criteria:

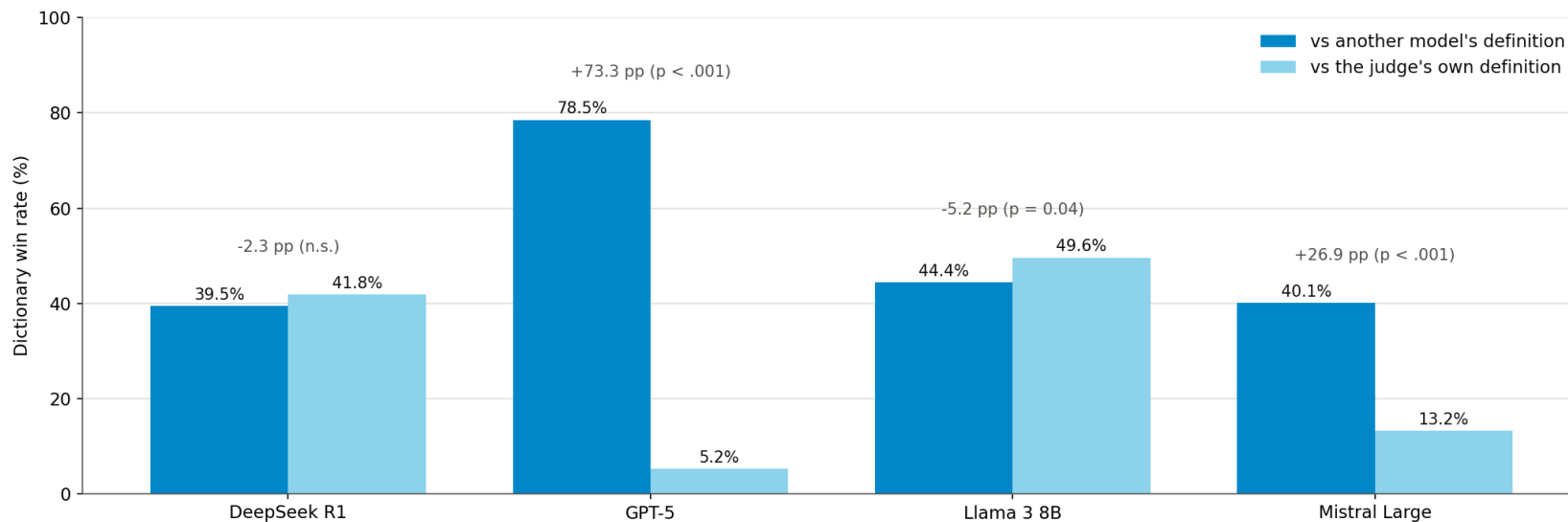
"You are the editor of an English language dictionary. Two writers have proposed definitions for a given word. It is up to you to decide which definition to include in the dictionary."

The definition shown first wins 65.7% of the time



RESULT 2 · SELF-PREFERENCE

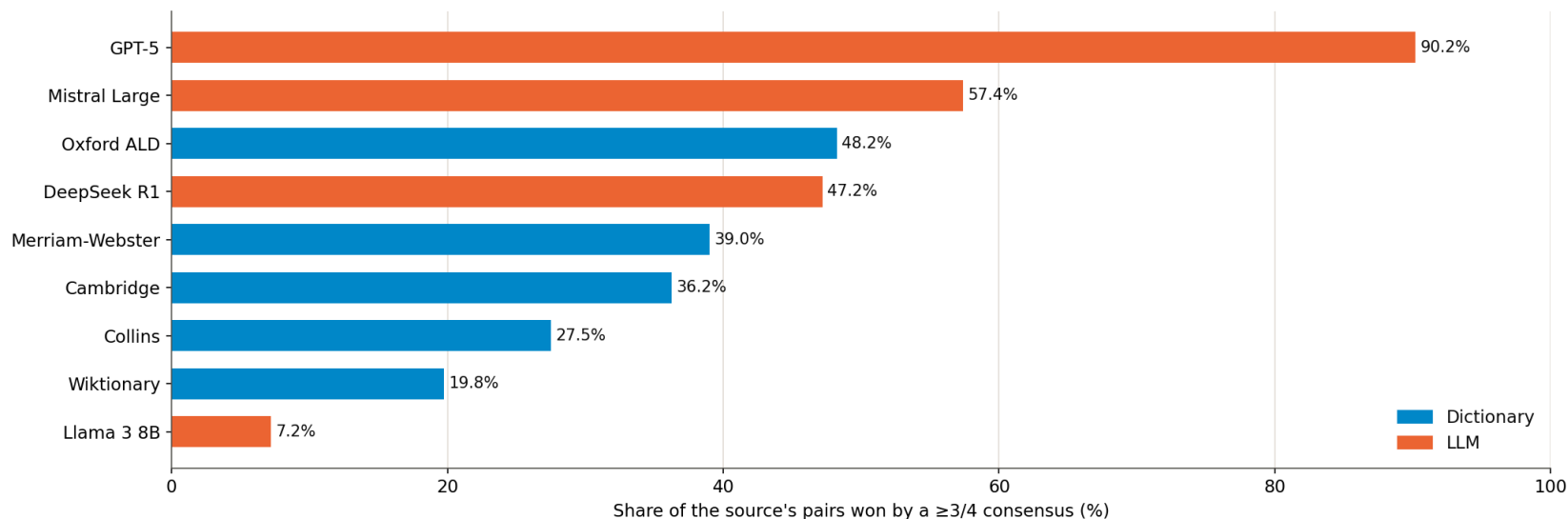
The strongest model is the most self-preferring judge



RESULT 3 · CONSENSUS

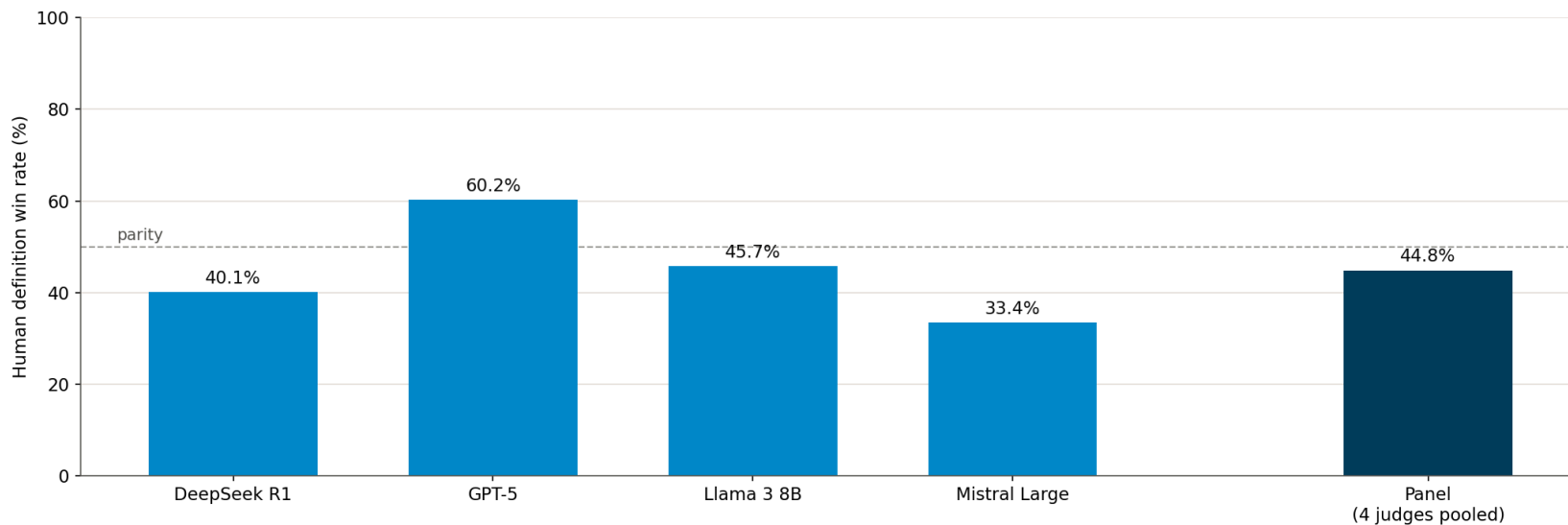
At committee level, LLM definitions win half of all pairs

LLM definitions 50.5% of pairs · dictionaries 34.2% · no consensus 15.3%



RESULT 4 · WIN RATES

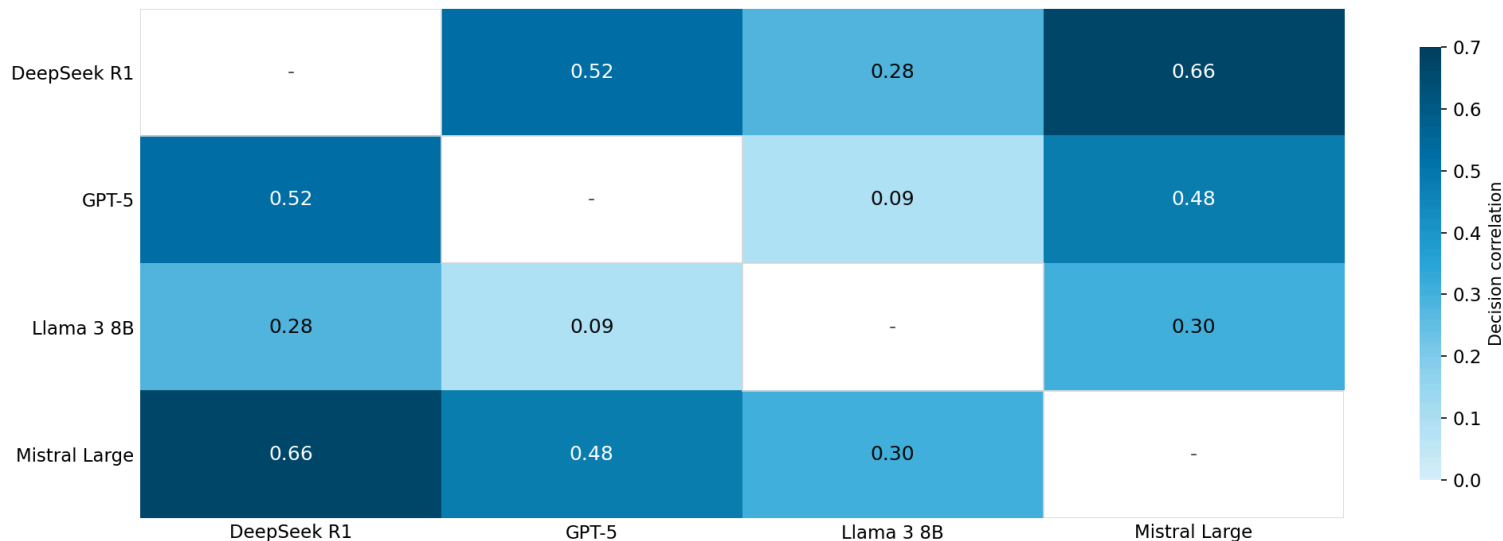
Opposite individual biases average out in the panel



RESULT 5 · INTER-JUDGE AGREEMENT

Judges barely agree with each other

Fleiss' κ = 0.357 (moderate) · pairwise decision correlation from 0.09 to 0.66



Limitations

- **Scope:** 100 high-frequency English words, one genre (lexicographic definitions)
 - **No human control group:** we measure disagreement between judges, not accuracy
 - **One prompt, one run per pair:** prompt sensitivity and self-consistency untested
 - **4 judges:** enough to show the effect, too few to model it
 - **Consensus $\neq$ ground truth:** converging judges can converge on the wrong answer
-

These limits bound the **magnitude** of our numbers - not the **existence** of the biases. Position and self-preference are measured on identical content.

CONCLUSION

Sophistication $\neq$ impartiality

Narrow, controlled setting: curated human references, blind protocol

Biases still massive

65.7% position - the definition shown first wins (3/4 judges)

+73.3 pp self-preference, at worst

55.2% AI-AI - the panel's preference for synthetic text

A judge not perceived as neutral will not be trusted - and will not be adopted

01

Measure the biases of your judges

02

Deploy **blind, diverse panels**

03

Calibrate continuously against human ground truth

Thank You!

Questions?

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